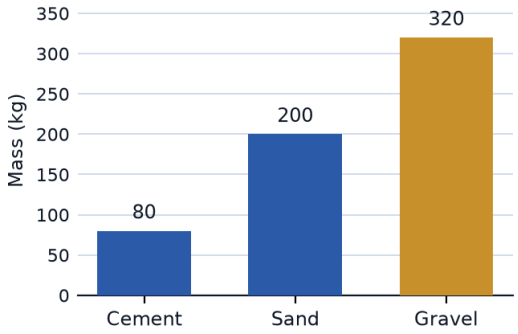
# Problem-Solving, Ratios, Rates & Data

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • [FRQ] • Target: Multi-Part Ratios & Proportional Partitioning

In a concrete mix, the ratio of cement to sand to gravel by weight is $1 : 2.5 : 4$. How many kilograms of each material are needed to produce $600\,\text{kg}$ of concrete?




**2** [Foundational] • [FRQ] • Target: Dimensional Analysis & Unit Conversion

Convert a speed of $90\,\text{km/h}$ into meters per second ($\text{m/s}$) and feet per second ($\text{ft/s}$, using $1\,\text{m} \approx 3.28\,\text{ft}$).

**3** [Foundational] • [FRQ] • Target: Successive Percentage Changes

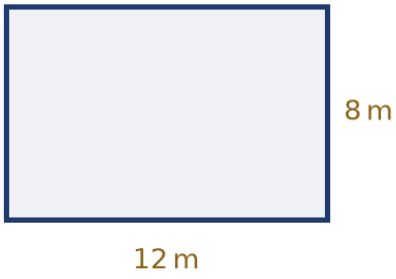
An electronics store marks up the wholesale cost of a laptop by $40\%$ to set the retail list price, and then offers a $25\%$ holiday discount off the list price. Find the net percentage profit over wholesale cost.

**4** [Foundational] • [FRQ] • Target: Reverse Percentage Problems

After a $16\%$ price reduction, a laboratory microscope sells for $630$ dollars. What was the original price of the microscope before the reduction?

**5** [Foundational] • [FRQ] • Target: Scale Drawings: Linear vs. Area Scaling

On an architectural blueprint drawn at a scale of $1\,\text{cm} : 2.5\,\text{m}$, a rectangular conference room measures $4.8\,\text{cm}$ by $3.2\,\text{cm}$. Find the actual perimeter (in meters) and actual floor area (in $\text{m}^2$) of the room.



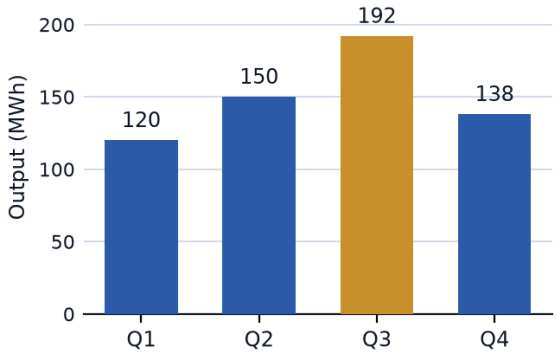

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**6** [Foundational] • [FRQ] • Target: Combined Work Rates

Pump A can fill a reservoir in $6$ hours, and Pump B can fill the same reservoir in $10$ hours. If both pumps operate together, how many hours does it take to fill the reservoir?

**7** [Foundational] • [FRQ] • Target: Reading Rates & Percentages from Bar Charts

The bar chart below shows quarterly solar energy output (in MWh) for a microgrid across four quarters. Calculate the percentage increase from Q1 to Q3 and the percentage of annual output generated in Q3.




## Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

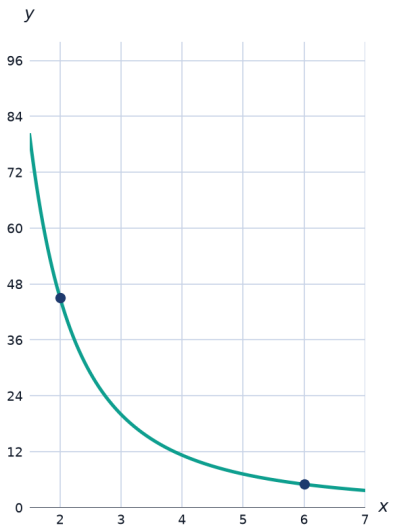
**8** [Intermediate] • [FRQ] • Target: Concentration Enrichment Equations

A metallurgist has $80\,\text{g}$ of a $35\%$ silver alloy. How many grams of pure ($100\%$) silver must be melted into the alloy to raise the silver concentration to $60\%$?

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**9** **[Intermediate]** • **[FRQ]** • **Target: Inverse-Square Variation & Percentage Change**

The illumination intensity $I$ from a point light source varies inversely as the square of the distance $d$ from the source ($I = k/d^2$). If $I = 180\,\text{lux}$ at $d = 2\,\text{m}$, find $I$ at $d = 6\,\text{m}$, and determine by what percentage $I$ changes when $d$ increases by $25\%$.




**10** **[Intermediate]** • **[FRQ]** • **Target: Weighted Average Speed Across Segments**

A commuter drives the first $60\,\text{km}$ of a trip at $80\,\text{km/h}$ and the remaining $90\,\text{km}$ at $60\,\text{km/h}$. Compute the total travel time and the overall average speed for the $150\,\text{km}$ trip.

**11** **[Intermediate]** • **[FRQ]** • **Target: Percentage Inclusion-Exclusion**

In a student body of $1{,}200$ students, $65\%$ take Spanish, $45\%$ take French, and $20\%$ take neither language. How many students take both Spanish and French, and what percentage of Spanish students also take French?

**12** **[Intermediate]** • **[FRQ]** • **Target: Multi-Step Rate Unit Conversion**

Convert an irrigation flow rate of $450$ gallons per hour into liters per minute, given $1\,\text{gallon} = 3.785\,\text{L}$. Round to two decimal places.

**13** **[Intermediate]** • **[FRQ]** • **Target: Compound Percentage Recovery**

A stock portfolio drops by $20\%$ in Year 1 and drops by another $10\%$ in Year 2. By what percentage must the portfolio grow in Year 3 to return to its exact initial value?

**14** **[Intermediate]** • **[FRQ]** • **Target: Chained Ratios & Homogeneous Expressions**

If $a : b = 3 : 4$ and $b : c = 6 : 5$, find the compound ratio $a : b : c$ in lowest integers and evaluate $\frac{2a + 3c}{4b}$.

## Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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**15** [Challenging] • [FRQ] • Target: Invariant Dry-Matter Percentage Problems • Cross-Lesson: Topic 1.01 (Conservation Linear Equations)

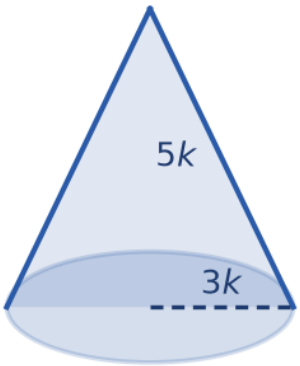
Fresh berries contain $80\%$ water by weight, whereas dried berries contain $20\%$ water by weight. How many kilograms of dried berries are obtained from $50\,\text{kg}$ of fresh berries?

**16** [Challenging] • [FRQ] • Target: Power-Law Proportionality & Percentage Scaling • Cross-Lesson: Topic 4.01 (Fractional Exponents)

According to Kepler's Third Law, the square of a planet's orbital period $T$ is directly proportional to the cube of its semi-major axis $a$ ($T^2 = ka^3$). If Planet X has a semi-major axis that is $4$ times that of Earth ($a_E = 1\,\text{AU}, T_E = 1\,\text{yr}$), find Planet X's orbital period and the percentage increase in $T$ when $a$ increases by $21\%$.

**17** [Challenging] • [FRQ] • Target: Similarity Scaling of 2D Area & 3D Volume • Cross-Lesson: Topic 8.01 (3D Solids & Similarity)

Two similar right circular cones have surface areas in the ratio $9:25$. Find (a) the ratio of their heights, (b) the ratio of their volumes, and (c) the percentage by which the larger cone's volume exceeds the smaller cone's volume.




**18** [Challenging] • [FRQ] • Target: Relative Risk, Sensitivity & Bayesian Proportions • Cross-Lesson: Topic 6.01 (Conditional Probability & Two-Way Tables)

A medical screening test for a condition present in $2\%$ of a population has a sensitivity (true-positive rate) of $95\%$ and a specificity (true-negative rate) of $90\%$. Out of $10{,}000$ people tested, how many test positive in total, and what percentage of those who test positive actually have the condition (Positive Predictive Value)?

**19** [Challenging] • [FRQ] • Target: Repeated Percentage Decay & Exponential Inequalities • Cross-Lesson: Topic 4.01 (Exponential Decay) & Topic 2.01 (Inequalities)

Under continuous-rate radioactive decay, a sample loses $10\%$ of its remaining mass every hour. Write an inequality for the number of hours $t$ until less than $50\%$ of the initial mass remains, and determine the smallest integer hour $t$ that satisfies it.

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**20** [Challenging] • [FRQ] • Target: Quadratic Revenue Elasticity from Percentage Changes • Cross-Lesson: Topic 3.01 (Quadratic Functions & Roots)

An airline increases its ticket price by $p\%$ and observes that ticket sales volume decreases by $\frac{p}{2}\%$. Express the percentage change in total revenue as a quadratic function of $p$, and find the non-zero value of $p$ for which total revenue remains unchanged.

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## Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

**21** [Foundational] • [MCQ] • Target: Simplifying Algebraic Ratios

If $\frac{3x}{5y} = \frac{9}{10}$, what is the ratio of $x$ to $y$?

**a.** $2:3$

**b.** $3:2$

**c.** $9:50$

**d.** $27:50$

**22** [Foundational] • [MCQ] • Target: Discount & Tax Calculations

A shirt originally priced at $60$ dollars is on sale for $25\%$ off. If an $8\%$ sales tax is applied to the sale price, what is the final amount paid?

**a.** $45.00$ dollars

**b.** $48.60$ dollars

**c.** $49.80$ dollars

**d.** $51.00$ dollars

**23** [Foundational] • [MCQ] • Target: Unit Rates & Time Conversion

A 3D printer produces $18$ identical gears in $45$ minutes. At this constant rate, how many gears can it produce in $2.5$ hours?

**a.** $45$

**b.** $54$

**c.** $60$

**d.** $75$

**24** [Foundational] • [MCQ] • Target: Equating Percentage Expressions

If $40\%$ of $x$ is equal to $16\%$ of $150$, what is the value of $x$?

**a.** $24$

**b.** $48$

**c.** $60$

**d.** $96$

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**25** [Foundational] • [MCQ] • Target: Direct Proportions & Map Scales

A map uses a scale of $1$ inch $= 15$ miles. If two cities are $4.6$ inches apart on the map, what is the actual straight-line distance between them?

**a.** 60 miles

**b.** 64 miles

**c.** 69 miles

**d.** 75 miles

**26** [Foundational] • [MCQ] • Target: Percentage Change Formula

The price of a commodity increased from $80$ dollars to $100$ dollars. What was the percentage increase?

**a.** 20%

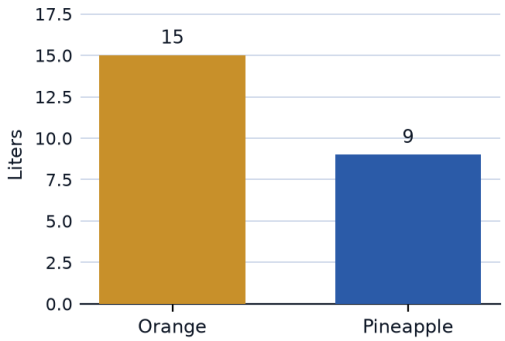
**b.** 25%

**c.** 80%

**d.** 125%

**27** [Foundational] • [MCQ] • Target: Part-to-Whole Ratio Calculations

A fruit punch recipe calls for orange juice and pineapple juice in a ratio of $5 : 3$. How many liters of orange juice are in $24$ liters of punch?




**a.** 9

**b.** 12

**c.** 15

**d.** 18

**28** [Foundational] • [MCQ] • Target: Direct Variation

If $y$ is directly proportional to $x$, and $y = 18$ when $x = 12$, what is the value of $y$ when $x = 20$?

**a.** 24

**b.** 26

**c.** 30

**d.** 32

## Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

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**29** [Intermediate] • [MCQ] • Target: Compound Percentage Change in Area

The length of a rectangle is increased by $20\%$ and its width is decreased by $15\%$. How does the area of the rectangle change?

**a.** It increases by $5\%$.

**b.** It increases by $2\%$.

**c.** It decreases by $2\%$.

**d.** It remains unchanged.

**30** [Intermediate] • [MCQ] • Target: Inverse-Square Proportions

Sound intensity $I$ is inversely proportional to the square of distance $r$ from the speaker. If a listener moves from $r = 3\,\text{m}$ to $r = 6\,\text{m}$, the new intensity is what fraction of the original intensity?

**a.** $\frac{1}{16}$

**b.** $\frac{1}{4}$

**c.** $\frac{1}{3}$

**d.** $\frac{1}{2}$

**31** [Intermediate] • [MCQ] • Target: Evaporation & Concentration Ratios

How many liters of water must be evaporated from $50\,\text{L}$ of a $12\%$ salt solution to produce a $20\%$ salt solution?

**a.** $15\,\text{L}$

**b.** $20\,\text{L}$

**c.** $25\,\text{L}$

**d.** $30\,\text{L}$

**32** [Intermediate] • [MCQ] • Target: Algebraic Reverse Percentages

After a $20\%$ discount, the sale price of a coat is $p$ dollars. Which expression represents the original price of the coat before the discount?

**a.** $0.80p$

**b.** $1.20p$

**c.** $1.25p$

**d.** $\frac{p}{1.20}$

**33** [Intermediate] • [MCQ] • Target: Updating Component Ratios

An alloy consists of copper and zinc in a $7:3$ ratio by mass. If $6\,\text{kg}$ of zinc is added to $40\,\text{kg}$ of this alloy, what is the new ratio of copper to zinc?

**a.** $7:4$

**b.** $14:9$

**c.** $7:5$

**d.** $2:1$

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**34** [Intermediate] • [MCQ] • Target: Density, Mass & Volume Unit Conversions

Titanium has a density of $4.5\ \text{g/cm}^3$. What is the mass, in kilograms, of a solid titanium block with volume $2{,}000\ \text{cm}^3$?

**a.** $0.9\ \text{kg}$

**b.** $4.5\ \text{kg}$

**c.** $9.0\ \text{kg}$

**d.** $90\ \text{kg}$

**35** [Intermediate] • [MCQ] • Target: Opposing Rates (Fill vs. Drain)

Pipe A can fill a tank in $4$ hours, while a drain valve B can empty the full tank in $6$ hours. If both Pipe A and drain valve B are accidentally left open when the tank is empty, how many hours will it take to fill the tank?

**a.** $8$

**b.** $10$

**c.** $12$

**d.** $24$

**36** [Intermediate] • [MCQ] • Target: Chained Percentage Multipliers

If $a$ is $25\%$ greater than $b$, and $b$ is $20\%$ less than $c$, then $a$ is what percent of $c$?

**a.** $95\%$

**b.** $100\%$

**c.** $105\%$

**d.** $110\%$

**37** [Intermediate] • [MCQ] • Target: Harmonic Mean for Equal-Distance Round Trips

A cyclist rides uphill at $12\ \text{km/h}$ and immediately rides back down the exact same trail at $36\ \text{km/h}$. What is the cyclist's average speed for the entire round trip?

**a.** $18\ \text{km/h}$

**b.** $20\ \text{km/h}$

**c.** $24\ \text{km/h}$

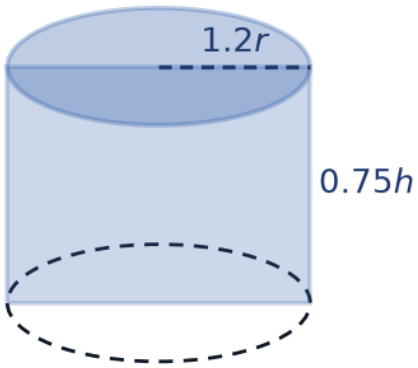
**d.** $27\ \text{km/h}$

## Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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**38** [Challenging] • [MCQ] • Target: Multivariable Scaling of 3D Cylinder Volume • Cross-Lesson: Topic 8.01 (Volume of 3D Solids)

If the radius of a right circular cylinder is increased by $20\%$ and its height is decreased by $25\%$, by what percentage does the volume of the cylinder change?




**a.** Decreases by $5\%$

**b.** Increases by $8\%$

**c.** Increases by $10\%$

**d.** Decreases by $10\%$

**39** [Challenging] • [MCQ] • Target: Drain-and-Replace Mixture Modeling • Cross-Lesson: Topic 1.01 (Linear Conservation Equations)

A $10$-liter radiator is filled with a $60\%$ antifreeze solution. How many liters of the solution must be drained and replaced with pure ($100\%$) antifreeze so that the resulting $10$-liter mixture is $75\%$ antifreeze?

**a.** $2.5\,\text{L}$

**b.** $3.25\,\text{L}$

**c.** $3.75\,\text{L}$

**d.** $4.0\,\text{L}$

**40** [Challenging] • [MCQ] • Target: Compound Percentage Extrapolation • Cross-Lesson: Topic 4.01 (Exponential Growth)

An investment grows at a constant annual compound rate so that its value increases by $44\%$ over $2$ years. What is the percentage increase over $3$ years?

**a.** $66\%$

**b.** $72.8\%$

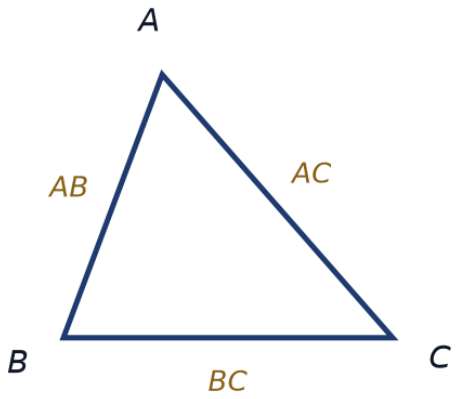
**c.** $88\%$

**d.** $95.2\%$

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**41** [Challenging] • [MCQ] • Target: Area Ratios in Similar Triangles • Cross-Lesson: Topic 7.01 (Similar Triangles)

In $\Delta ABC$, point $D$ lies on $AB$ such that $AD : DB = 2 : 3$, and point $E$ lies on $AC$ such that $DE \parallel BC$. The area of $\Delta ADE$ is what percent of the area of trapezoid $DBCE$?




**a.** 16%

**b.** 19.05%

**c.** 40%

**d.** 66.7%

**42** [Challenging] • [MCQ] • Target: Symmetric Percentage Rise & Fall via Quadratics • Cross-Lesson: Topic 3.01 (Difference of Squares)

A store raises the price of an item by $x\%$ and later reduces the new price by $x\%$, resulting in a final price that is $9\%$ less than the original price. What is the value of $x$?

**a.** 9

**b.** 15

**c.** 30

**d.** 45

**43** [Challenging] • [MCQ] • Target: Ratio Rebalancing & Transfer • Cross-Lesson: Topic 2.01 (Linear Systems)

In a survey of $500$ voters, the ratio of voters favoring Candidate A to Candidate B is $7 : 3$. How many voters must switch their support from Candidate A to Candidate B so that the new ratio becomes $1 : 1$?

**a.** 50

**b.** 75

**c.** 100

**d.** 125

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**44** [Challenging] • [MCQ] • Target: Gear Ratios & Angular Velocity in Radians • Cross-Lesson: Topic 9.01 (Circles & Radian Measure)

Two gears are meshed together. Gear A has $36$ teeth and rotates at $150$ rpm. If Gear B has $24$ teeth, through how many radians does Gear B rotate in $2$ seconds?

**a.** $10\pi$

**b.** $15\pi$

**c.** $20\pi$

**d.** $30\pi$

**45** [Challenging] • [MCQ] • Target: Linear Invariance of Equal Proportions • Cross-Lesson: Topic 10.01 (Rational Expressions)

To satisfy the synthesis condition, if $\frac{a}{b} = \frac{c}{d} = \frac{3}{4}$ with $b + 2d \neq 0$, what is the value of $\frac{a+2c}{b+2d}$?

**a.** $\frac{3}{8}$

**b.** $\frac{3}{4}$

**c.** $\frac{9}{16}$

**d.** $\frac{6}{7}$

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